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PI Name:	Fuller, Charles A. Ph.D.		
Project Title:	Vestibular Modulation of the Circadian Timing System in Altered Gravity Environments		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
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Space Biology Special Category:	None		
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Comments:			
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Contact Monitor:		**Contact Phone:**	
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Task Description:	Organisms have evolved in Earth's static gravitational environment. Thus, it is perhaps not surprising that exposure to non-Earth gravitational fields can influence an organism's physiology and behavior. One of the affected physiological systems is the Circadian Timing System (CTS). The CTS is responsible for coordinating an organism's physiology and behavior, both internally and with the external 24-hour day. Proper CTS function is essential for maintaining normal physiology and behavior. Recent work in our laboratory has revealed that the vestibular system plays a role in mediating the CTS responses to changes in gravity. This program will elucidate the physiological and molecular mechanisms by which the vestibular system influences the CTS in an altered gravity environment. Our overall objective is to determine if the changes in circadian function induced by exposure to a hyperdynamic environment (i.e. 2G) are mediated by input from the vestibular sensory receptors located in the inner ear, specifically the macular receptors. In order to accomplish this objective, this task will elucidate the physiological and molecular influence of the vestibular sensors on the regulation of the endogenous circadian period in +G environments. In addition, we will determine the role of vestibular inputs on the regulation of the physiological and molecular responses of the circadian pacemaker to photic stimuli in +G environments. The circadian timing system (CTS) is essential for the generation and internal temporal coordination of biochemical, physiological and behavioral rhythms, as well as for synchronizing these physiological oscillations with the external 24-hour environment. When CTS function is disrupted, an organism's ability to respond appropriately to a physiological stressor is inhibited. Patients with CTS dysfunction often suffer from sleep-wake, affective and neuroendocrine disorders. In addition, CTS dysfunction can severely compromise an individual's long-term health. The vestibular system is responsible for perception, orientation and equilibrium within the three dimensional environment. Over the last 10 years a wider functional role has been demonstrated for the vestibular system. The vestibular system can modulate cardiovascular and respiratory function and has a role in anxiety disorders and spatial navigation. We have recently described a role for the vestibular system in the regulation of physiological homeostasis and circadian function. Our objectives are aimed at understanding how the vestibular system mediates CTS responses to changes in gravity. Vestibular mediated alterations in the period of the pacemaker would affect the ability of the CTS to synchronize with the external environment. Any influence of the vestibular system on photic responsiveness of the CTS would have important implications for both astronauts in altered gravity environments as well as individuals with vestibular and/or circadian disorders.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although considerable knowledge concerning the physiology of the vestibular system and the Circadian Timing System (CTS) has been gained over the past 10 years, little is known about the interaction of the two systems in altered gravity. The proposed studies will reveal fundamental aspects of the physiology of both systems. Circadian rhythms are fundamental regulatory processes of the nervous system and, as such, the putative ability of the vestibular system to modulate the CTS is of great potential significance with clinical and basic science applications. By understanding the physiological and molecular mechanisms by which the vestibular and circadian systems interact, therapeutic and diagnostic strategies may be developed to alleviate vestibular circadian and sleep-wake disorders with vestibular and/or circadian bases. Because of the ubiquitous endogenous control the CTS has on physiology and behavior, this research has a high degree of relevance to a number of disorders, including ones with no prior linkage to vestibular function.
Task Progress:	Vestibular function is critical to sensorimotor systems that provide an organism with the means to maintain postural equilibrium, sensory orientation and spatial perception. In addition, our recent evidence has shown that the vestibular system plays a role in neurobiological and physiological regulatory systems, including the circadian timing system. Therefore, this program contrasted the responses of the CTS of normal mice (PWT) with those of a genetic mutant, the head tilt, or het, mouse. This mouse has a spontaneous, homozygous mutation on chromosome 17 that results in a lack of otoconia within the macular gravity receptors. The auditory and rest of the labyrinth appears normal. Het mice appear to have normal hearing and respond normally to cranial rotation. The period of the circadian clock is affected by light intensity. Therefore, the period of the body temperature and activity rhythms of het and PWT mice were determined under three lighting conditions: constant darkness (DD), constant dim light (LLD) and constant bright light (LLB). In DD, the period of the Tb rhythm was significantly longer ($p<0.001$) in het than in PWT mice. In contrast, the daily mean Tb and Tb circadian rhythm amplitude did not differ between het and PWT mice. Similar to Tb, in DD, the het mice demonstrated a significantly longer ($p<0.001$) circadian period of ACT as compared to PWT mice. Again, mean daily ACT and ACT circadian rhythm amplitude, in DD, were similar between the het and PWT mice. In LLB, the het mice demonstrated a significantly shorter ($p<0.01$) circadian period of Tb compared to the PWT mice. The circadian period of Tb was significantly longer ($p<0.01$) in LLB than in DD for both the het and PWT groups. However, increasing ambient illuminance (0 µmoles s-1 m-2 to 0.5 µmoles s-1 m-2) had a significantly greater ($p<0.01$) period-lengthening effect in the PWT group than in the het group. Similar to Tb, in LLB, the circadian period of ACT was also significantly longer ($p<0.01$) in LLB than in DD for both the het and PWT groups. Also similar to Tb, the increased light intensity (DD to LLB) had a significantly greater ($p<0.01$) period-lengthening effect on ACT in the PWT group compared to the het group. In LLD only the PWT mice demonstrated a significant change ($p<0.01$) in the circadian period of Tb and ACT. Similar to DD, mean daily Tb, Tb circadian amplitude, daily mean ACT and ACT circadian amplitude did not differ between groups in either LLB or LLD. The PWT and het mice groups demonstrated quantitatively equivalent negative light masking (i.e., suppression) of ACT when comparing data from DD and LLB. Here, negative light masking was assessed as a % drop in mean ACT levels in LL as compared to DD. Moreover, the het and PWT mice demonstrated ACT alpha (activity) to rho (rest) ratios in DD, in LLB and in LLD similar to that observed in LD, i.e., greater activity consolidation in the het mice. The magnitude of the decrease in the ACT alpha/rho ratio between DD and both LLB and LLD was similar between het and PWT mice, however. Collectively, these observations suggest that enhanced negative light masking does not explain the changes in activity consolidation seen in the het mice. Prior to release into constant darkness, under a 24-hour LD cycle (LD 12:12) daily mean Tb and Tb circadian rhythm phase and amplitude did not differ between het and PWT mice. Daily mean ACT and ACT circadian rhythm phase and amplitude also did not differ between het and PWT mice. Despite equivalent mean ACT levels, however, the distribution of activity between subjective night and day was different between het and PWT mice. Specifically, during LD, the het mice were

	significantly less active during the subjective day (16.5+5.1%) than PWT mice (31+6%), suggesting possible differences in behavioral state control, e.g., increased sleep in het mice. The present study demonstrates that mice lacking otoconia, and hence the perception of inertial and gravity accelerations, exhibit alterations in the period of the intrinsic circadian pacemaker and the period-lengthening response to changes in tonic light intensity. These results support the proposal that inputs from the vestibular system can modulate the CTS and represent the first neurobehavioral evidence for a vestibular-CTS interrelationship. As such, these data suggest multi-sensory regulation of the CTS as well as a possible novel vestibular etiology for a host of circadian-related disorders. As an example, because the intrinsic period of the circadian pacemaker is a determinant of the phase angle of entrainment, vestibular-mediated alterations in the period of the pacemaker could affect the ability of the CTS to synchronize with the external environment, i.e., entrainment. CTS dysfunction has also been implicated in sleep-wake, psychiatric, and aging-related disorders, e.g., familial advanced sleep phase syndrome, whose sufferers possess a mutation in a key clock gene, per2, which alters the intrinsic period of the circadian pacemaker. More generalized circadian dysfunction, with or without an altered period basis, has been demonstrated to decrease homeostatic capacity and linked with increased cancer incidence. In summary, the results of the present study: 1) provide the first neurobehavioral evidence for a vestibular-CTS interrelationship; 2) contribute to our understanding of the regulation of the intrinsic circadian period; 3) suggest a novel afferent role for the macular gravity receptors in transducing activity-based non-photic stimuli; 4) may have important implications for space-bound individuals; and, perhaps most importantly, 5) have practical implications for understanding the pathophysiology of, and developing treatments for, circadian and circadian sleep-wake disorders with vestibular etiologies.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Articles in Peer-reviewed Journals	Fuller PM, Fuller CA. "Genetic evidence for a neurovestibular influence on the mammalian circadian pacemaker." J Biol Rhythms. 2006 Jun;21(3):177-84. PMID: 16731657 , Jun-2006